higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 2

(12020162)

22 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only a BLUE or BLACK pen.
6. Write neatly and legibly.

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QUESTION 1: INTRODUCTION TO PLUMBING

1.1 Why were plumbing systems like water carrying pipes and drains developed? (2)

1.2 Give TWO future developments in the plumbing industry. (2)

1.3 Give THREE main properties of pure water. (3)

1.4 Briefly explain the following terms:

1.4.1 Water cycle

1.4.2 Screening

(2 × 2) (4)

1.5 List FOUR roles a plumbing manager must fulfil. (4)

[15]

QUESTION 2: PLUMBING PRINCIPLES & CODE OF PRACTICE

2.1 Briefly explain the following terms:

2.1.1 Surface water

2.1.2 Ground water

(2 × 2) (4)

2.2 Give TWO examples of ground water. (2)

2.3 Name THREE systems offer certificates of approval. (3)

2.4 What is the formula to calculate the pressure of water at a point? (2)

2.5 Give TWO ways of causing water to flow through pipes. (2)

2.6 Describe the term *flow rate.* (2)

[15]

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QUESTION 3: MATERIALS, TOOLS & EQUIPMENT

3.1 Choose a description from COLUMN B that matches an item in COLUMN A. Write only the letter (A–F) next to the question number (3.1.1–3.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1.1	A scriber	A	clearing waste that is blocking pipes
3.1.2	Basin wrench	B	use to mark circles and arcs
3.1.3	Plunger	C	cut plastic and copper pipe & gms
3.1.4	Hacksaw	D	to reach tight spots under sinks and basins
3.1.5	Divider	E	check if walls are plumb
		F	a sharp instrument used to mark lines when you are measuring

(5 × 1) (5)

3.2 Give TWO examples of waste water fixtures. (2)

3.3 Explain the term w*ater hammer.* (2)

3.4 Name a type of ladder that can be used in plumbing. (1)

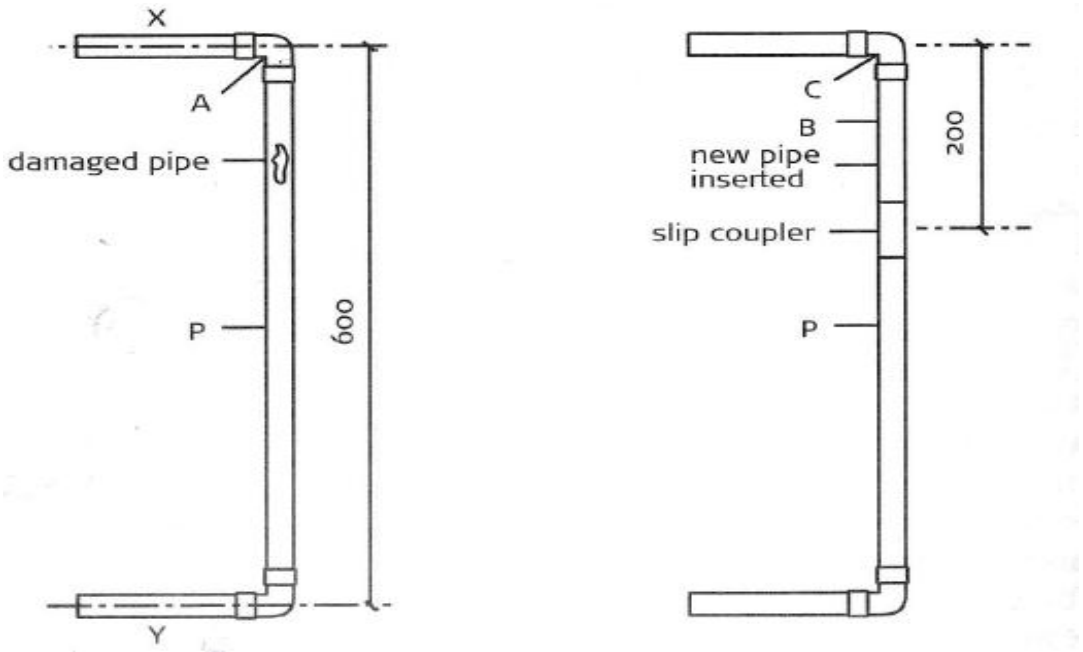
[10]

QUESTION 4: WATER PIPE ASSEMBLY

4.1 List FIVE tools/equipment you would require to use, measure, cut, thread and assemble GMS pipe. (5)

4.2 Give TWO ways in which you can join copper pipe. (2)

4.3 The section of the pipe marked B in the drawing is the section of pipe that needs to be replaced. The pipe work marked X & Y must be left in the place and not removed from the wall.



Explain the procedure to fix the damaged pipe? (7)

4.4 Briefly explain the term *crimping.* (2)

4.5 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.5.1–4.5.5) in the ANSWER BOOK.

4.5.1 Solvent cement or PVC weld is a type of glue used to join PVC pipes.

4.5.2 Mono means two.

4.5.3 Polyethelyne is light but strong and is used for making cold water pipes and cisterns.

4.5.4 Polypropylene is more expensive than other pipes.

4.5.5 A pipe wrench is used to cut copper pipes and galvanised pipes.

(5 × 1) (5)

4.6 List FOUR methods to bend a multilayer pipe. (4)

[25]

QUESTION 5: CUTTING AND JOINING OF METALS

5.1 What is the most common hand tool to cut sheet metal? (1)

5.2 Briefly explain what is meant by the term *flashing?* (2)

5.3 Give ONE word or a term for each of the following descriptions by choosing a word or term from the list below. Write only the word or term next to the question number (5.3.1–5.3.5) in the ANSWER BOOK.

soakers; skirting; zinc; mitre; 10 mm; vent pipe; quadrant

5.3.1 In roofing … is the portion of the roofing that is turned up against a vertical surface, like wall or chimney face.

5.3.2 … are short pieces of material that is used with stepped flashing.

5.3.3 … is used to coat mild steel to prevent rusting.

5.3.4 … is when something is cut at an angle.

5.3.5 To finish the job you will need to stagger the bottom edge of cone with alternate … laps.

(5 × 1) (5)

5.4 Name TWO methods that can be used to join sheet metal. (2)

[10]

QUESTION 6: SET OUT BELOW GROUND DRAINAGE SYSTEMS

6.1 List SIX main points to consider when laying out a drainage system. (6)

6.2 What is s gradient on a plumbing plan? (1)

6.3 Draw a neat labelled sketch, indicating the crown and the invert of a drainage pipe. (3)

6.4 Describe the meaning of the following symbols:

6.4.1 

6.4.2 

6.4.3 

6.4.4 

6.4.5 

(5 × 1) (5)

6.5 Briefly explain the term *manhole.* (2)

6.6 Name TWO types of materials used for bedding. (2)

6.7 What is the function of a temporary timbering? (2)

6.8 List FOUR other methods you could use to check the levels and gradient of a drain in a trench. (4)

[25]

TOTAL: 100

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